ICDEC

Final Report Template

Intern Name		**Topic**	
Supervisor		**Status**	Draft
Target Hardware	Grove Vision AI V2 (Cortex-M55 + Ethos-U55)		

Guidance text in shaded boxes is instructional — delete it once each section is filled in. Flexible sections (signal processing detail, ablation, live monitoring) are optional; add them as extra numbered sections after Section 7 if relevant to your topic.

1. Overview

Topic, objective, sensor used, target hardware. 3-5 sentences.

2. Background

Why this problem matters, domain-relevant metrics (e.g. what counts as clinically useful accuracy, standard benchmarks in the literature). Keep to the essentials that justify your design choices later — this isn't a literature review.

3. Dataset & Preprocessing

Dataset name/source, size, subjects, sampling rate, labels, split (e.g. LOSO), class balance if relevant. Preprocessing steps: filtering, windowing, normalization, augmentation. Note what runs in firmware C on-device vs. Python-side only during training.

Optional: signal processing detail — filter design, windowing strategy, feature extraction, if it involved meaningful design decisions.

4. Model

Architecture (layer-by-layer or diagram), parameter count, input shape, training setup: optimizer, loss, epochs, batch size, training hardware, regularization.

5. Results

Accuracy/metric table for your task, comparison to a baseline if available, error analysis (confusion matrix, failure cases, per-class breakdown).

Optional: ablation study — which architectural or preprocessing choices actually mattered. Only include if you ran one.

6. Deployment

6.1 Standard Metric Summary

Mandatory *— exact column names, one row per model variant deployed. Feeds directly into the NPU reference workload table.*

Sensor	Model Arch	Input Shape	Params	MAC	Flash	SRAM	Latency	Accuracy	Quantization

6.2 Deployment Notes

Quantization method (PTQ/QAT), toolchain used (Vela, ExecuTorch, etc.), NPU mapping/utilization %, memory footprint breakdown, on-device latency measurement method.

Optional: live monitoring — if a demo was built (live sensor feed → inference → display/output), describe setup and behavior here or as its own section.

7. Conclusion + Challenges

What worked, what didn't, what you'd do differently with more time, open issues for the next intern on this topic. One paragraph, direct.